

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)	Temperature change = 22.5 [°C] (1) Substitution: $5400 = 0.5 \times c \times 22.5$ (1) ecf on attempted temp change subtraction shown $c = 480$ [J/kg °C] (1)	1	1 1		3	2	3
		(ii)	Difference of {95 / around 100} (1) ecf {Not close / 25%} so not accurate (1) ecf			2	2	1	2
		(iii)	Make sure all of the heater is inside the block (1) OWTTE Wrap block in insulating material (1) Accept: use a more powerful heater use a balance with a better resolution add oil to the hole for the thermometer Don't accept use a thermometer with a better resolution			2	2		2
	(b)		[Atoms] gain [kinetic] <u>energy</u> (1) <u>Vibrate faster</u> (1) accept more Don't accept start to vibrate Treat reference to electrons as neutral	2			2		2
			Question 4 total	3	2	4	9	3	9